

How to Prepare Your Papers for the JIP

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Abstract: This document is a guide for preparing drafts to be submitted to the Journal of Information Processing (JIP) and for the final camera-ready manuscripts of papers to appear in the JIP that use L^AT_EX and special style files. Since this document itself is produced with these style files, it will help you to refer to its source file, which is distributed with these style files.

1. Introduction

The Information Processing Society of Japan (IPSJ) publishes Journal of Information Processing (JIP) as its flagship international journal. Thus far, the JIP adopted the landscape A4 format for publishing papers, but it has changed this format to the portrait A4 format because of many requests from authors. Corresponding to this format change, the JIP accepts the portrait A4 format for submitting papers.

Following with this change, we, the Editorial Board of the JIP, prepared a new style file for L^AT_EX. In this manuscript, we first describe the usage of the style file. The basic strategy of the new style file is to require no special knowledge of command usage that does not use standard L^AT_EX commands. Authors of a paper can use standard L^AT_EX commands to keep within the formatting restrictions of the paper, such as setting space pitches and margins. The guideline of the paper format will be described in Section 4. Since this manuscript itself also is written with the style file, we hope it will help with writing a paper.

The Editorial Board of the JIP has also prepared a “Dos and Don’ts” list of matters an author should consider while writing a paper. We have added the list to the latter portion of this manuscript. Please use the list as a checklist for preparing to submit a paper.

2. Flow from Submission to Publishing

2.1 Preparation

The JIP author’s kit including the L^AT_EX style files can be downloaded from the following URL:

<http://www.ipsj.or.jp/jip/submit/style.html>

The kit includes the following files:

- (1) ipsj.cls: style file for ipsj journals
- (2) ipsjdraft.sty: style for drafts to be submitted
- (3) ipsjpref.sty: style for the foreword
- (4) jsample.tex: source for the Japanese version of this guide
- (5) esample.tex: source of this guide
- (6) ipsjsort-e.bst: bibtex style (sorted)
- (7) ipsjunsrt-e.bst: bibtex style (unsorted)
- (8) bibsample.bib: sample of bibliographic data (Japanese)
- (9) ebibsample.bib: sample of bibliographic data (English)

Since the kit has variants corresponding to multiple platforms, including UNIX workstations, Windows (DOS), and Macintosh machines, an appropriate variant can be selected and unpacked on the target platform.

Since L^AT_EX 2_ε is required as an execution environment, please install it.

Regarding manuscripts written with Microsoft Word, a corresponding company will convert them into L^AT_EX. This means that the Microsoft Word format is used just a reference.

2.2 Draft Submission

First, generate a PDF file from your L^AT_EX source and style file under your L^AT_EX 2_ε environment and check that the generated PDF file can be read with the Adobe PDF reader. After that, register your email address into the Paper Review Management System (PRMS) through the following URL:

https://www.ipsj.or.jp/prms/author_pre_submit.do

The system will return an email including another URL for submitting your paper. The manual for submission via the PRMS is available at the following URL:

http://www.ipsj.or.jp/jip/submit/manual/e_manual.html

The JIP adopts double blind review, where reviewers of

The real author is the Editorial Board of JIP.

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your paper will not know your name, and you will not know theirs. To ensure that this is possible, the submitted draft version should not contain information about the authors.

2.3 Final Version

After you receive the notification of acceptance, revise your paper in accordance with the comments from the referees and add the required omissions from the draft, such as a biography, if any. The layout of figures and tables should be fixed. After that, check your paper again and again to completely remove description errors.

Send both the L^AT_EX file package and the hard copy to the IPSJ. The standard contents of the file package are .tex and .bbl. If you include PostScript files and/or special style files, add them into the package. Note that you must not split your source into multiple .tex files because it is hard for printers to access multiple files when they modify your source. Also, carefully make sure that the package contains all necessary files, especially special style files.

Details on the file transfer, including its destination and packaging method, will be provided to you by the IPSJ secretariat.

2.4 Proofreading, Typesetting, and Publishing

The IPSJ may change terms in your paper as per its standard, and the printing house may modify your source to make it fit the standard printing style. Even if they make no changes, the result printed at the printing house may be different from what you printed because of differences in the L^AT_EX execution environment. Therefore, the galley proofs of your paper will be sent to you so that you can check if those modifications and/or differences are acceptable. If not, correct errors with red ink. Note that this proofreading is not for correcting your errors, which should have been corrected before sending the final version.

Your paper will be typeset after errors you notify us about (if any) are corrected and will be published as part of the JIP.

3. Guide for Formatting a Paper

The JIP, as opposed to conference proceedings, has a traditional and stiff style. This makes the style files also stiff and strongly restricts customizability, which is one of the most useful features of L^AT_EX. For example, you must not change style parameters, such as \texheight. It is not easy to show which customizations are allowed, but the rule “Don’t tamper with it unless you are confident” should suffice.

Note that if you do something you should not, you may not have error messages but simply unattractive results.

The source file must use the following format. Underlined parts can be omitted from draft versions.

4. Configuration of Paper

The source file must use the following format. Underlined parts can be omitted from draft versions. Note that a

few additional commands, shown in A.1 of the Appendix, are available for a paper included in the Transactions.

\documentclass[JIP]{ipsj} or
\documentclass[JIP,draft]{ipsj}
 Specify other option styles if necessary.
 Specify auxiliary styles with \usepackage.

\setcounter{volume}{<volume>}
\setcounter{number}{<number>}
\setcounter{page}{<first-page>}
\received{<year>}{<month>}{<day>}
\accepted{<year>}{<month>}{<day>}
 Define your own macros if necessary.

If you cannot use the txfonts package, please do not use the following command

\usepackage[varg]{txfonts}
\makeatletter
\input{ot1txtt.fd}
\makeatother

\begin{document}
\title{<title>}
\affiliat{<affiliation-label>}{<affiliation>}

Declare current affiliation with \paffilabel if necessary.
\paffiliate{<affiliation-label>}{<affiliation>}

\author{1st-author}{affiliation-label}[E-mail]
\author{2nd-author}{affiliation-label}

\begin{abstract}
 <abstract>
\end{abstract}

\begin{keyword}
 <keyword>
\end{keyword}

\maketitle

\section{heading-of-1st-section}

 <main text>

Put acknowledgments here with the acknowledgment environment if any.

\begin{acknowledgment}
\end{acknowledgment}

\begin{thebibliography}{99}%9 or 99
\bibitem{1}
\bibitem{2}

`\end{thebibliography}`

Put appendices here following `\appendix` if any.

`\appendix`

`\section{heading-of-1st-section}`

`\begin{biography}`

`\profile{<1st-author>}{<biography-of-1st-author>}`

`\profile{<2nd-author>}{<biography-of-2nd-author>}`

`\end{biography}`

`\end{document}`

4.1 Option Style

The following six styles are available as optional arguments of the `\documentclass`. If the JIP option is not used, the program will use the standard Japanese paper style as the default.

(1) JIP For English documents

(2) draft For draft versions

(3) invited For invited papers

(4) sigrecommended For a paper recommended by a SIG

(5) technote For technical notes

(6) preface For the preface of an issue

Any combination of these options can be used.

If you use `\documentclass[JIP,draft]{ipsj}`, the “draft” option style will be applied. If you specify auxiliary style files with the `\usepackage`, you must include them in the file package when you send your final version to the IPSJ.

However, style files included in the L^AT_EX 2_ε standard distribution (e.g., `graphicx`) may be omitted. Note that style files may be incompatible with the style of the Journal Transaction.

4.2 Title, Author Names, etc.

Describe the title of your paper, author names and affiliations, and abstract using the commands and environment shown in Section 4. Then, perform `\maketitle` to automatically put them at the appropriate position. In the draft version, the title and abstract are automatically printed onto separate pages, while author names and affiliations are not printed in order to make your paper anonymous.

4.2.1 Title

The title specified with `\title` is made centered. Even if the title is too long to fit onto one line, an automatic line break is not performed. If your title is long, insert `\\` into the appropriate positions to break the lines. A multiple line title is first flushed left and then centered with respect to the widest line.

The title also appears in the header of odd numbered pages. If your title is too long, provide a shortened title for the header to `\title` as its optional argument as follows.

4.2.2 Author Name and Affiliation

When indicating the affiliation of each author with a label (first citation) and starting from the first author, by using `\affiliates`, numbered footnotes will be generated that show the affiliations. When several authors are affiliated

with the same organization, the affiliation needs to be indicated only once. For the author’s current affiliation, use `\paffiliate` and provide the label and affiliated organization as before. If the affiliated organization arguments are entered as current and a line break is inserted using `\\`, the author name will be automatically defined by `\author`. Immediately after the author’s name, enter the affiliation label and the author’s e-mail address.

Where there are several authors, repeating `\author` will generate additional authors in sequence (two authors, three authors, and so forth).

To add current affiliations or multiple affiliations, delineate the affiliate label using commas to include additional data.

4.2.3 Abstract

The abstract of your paper should only be used in the abstract environment.

4.2.4 Keywords

The keywords of your paper should be included as the content for the keyword environment.

5. Main part

5.1 Sectioning

L^AT_EX standard commands such as `\section` and `\subsection` are available for sectioning. The section heading of `\section` occupies two lines, while others are put into one line.

5.2 Fixed Baselines

Each page of the JIP is formatted with the double-column style. The printing tradition of double-column requires that a line in the left column and its neighbor in the right column have the same baseline. To meet this requirement, the style files carefully control the progression of baselines when a vertical space is inserted for section titles and so on.

5.3 Font Size

You will see that various size fonts are used in the printed result of your paper. Since these fonts are automatically and carefully chosen by the style files, you are free from the headache of selecting proper fonts. In fact, it is strongly recommended not to use font-size-changing commands such as `\large` and `\small` in the main text because they are quite harmful to retaining fixed baselines.

5.4 Itemizing

There is no special format for itemization. You can use the standard `enumerate`, `itemize`, `description` environment.

5.5 Footnotes

The command `\footnote` produces footnotes with reference marks such as ^{*1} and ^{*2}. When there is more than one footnote within a single page, please note that it is

^{*1} An example of footnote 1.

^{*2} An example of footnote 2.

```
\begin{figure}[tb]
<figure-body>
\caption{<caption>}
\label{...}
\end{figure}\
```

Fig. 1 Single column figure with caption explicitly broken by \\

necessary to run L^AT_EX twice to process them correctly. Moreover, it is sometimes preferable to separate a footnote and its mark into different columns. This can be achieved using the \footnotemark and \footnotetext commands. The footnote numbering produced by L^AT_EX is continuous throughout the paper; it does not restart on each new page.

5.5.1 Overfulls and Underfulls

The final result must be free from any overfulls. It is well known that almost all overfulls can be avoided with a little effort when writing sentences. For example, avoiding long in-text formulas and \verb is very effective. However, tricks using the flushleft environment, \\, or \linebreak are not recommended because they cause quite unattractive results.

For underfulls, you will conveniently get the following warning message,

```
* Underfull \hbox (badness 10000) detected
```

, by inserting \\ at the end of a paragraph. This message is also output when you use \\ just before a list-like environment, just before an \item, and at the end of the environment. Such underfulls cause unattractive empty lines and a flood of warnings that will hide important error messages.

5.6 Formulas

5.6.1 In-text Formulas

In-text formulas may be surrounded by any proper math-open/close pair, i.e. \$ and \$, \((and \), or \begin and \end for the math environment. Note that tall materials in in-text formulas, such as $\frac{a}{b}$ (\frac{a}{b}), are unattractive and will disarrange the baseline progression.

5.6.2 Displayed Formulas

Displayed formulas must not be surrounded by the pair \$\$\$. Instead, use the \[and \] pair or one of the environments displaymath, equation, or eqnarray. These commands/environments indent formulas (not centered) and keep fixed baselines as follows.

$$\Delta_l = \sum_{i=l+1}^L \delta_{pi}. \quad (1)$$

5.6.3 Eqnarray environment

For a sequence of two or more related formulas (equations), use the eqnarray environment to line them up at equal (or unequal) signs instead of \[\] or the equation environment.

5.6.4 Special Fonts

It is strongly recommended to use only standard L^AT_EX math fonts. Otherwise, you must report that you are using special fonts.

```
\begin{minipage}[t]{%
{0.5\columnwidth}}
\captionType{table}
\caption{...}
\ecaption{...}
\label{...}
\makebox[\textwidth]{c}{%
\begin{tabular}[t]{lcr}
\hline\hline
left&center&right\\
\hline
L1&C1&R1\\
L2&C2&R2\\
\hline
\end{tabular}}
\end{minipage}
```

Fig. 2 Contents of Table 1

Table 1 A table built by Fig. 2

left	center	right
L1	C1	R1
L2	C2	R2

```
\begin{figure}[tb]
<figure-body>
\caption{<caption>}
\label{...}
\end{figure}\
```

Fig. 3 Single column figure with caption explicitly broken by \\

5.7 Figures

A figure fit to one column is specified by the form shown in Fig. 1. Note that you must not specify the h option.

The \caption of a figure should be given below the figure body together with a \label command. A long caption will be automatically broken into two or more lines and centered with respect to the widest line. You can assist, however, with the line breaking by adding \\ to obtain a more beautiful result, especially for two-line captions, as shown in Fig. 1.

If you want to rank two or more figures and/or tables in a figure (or table) environment in order to save space, enclose each figure/table and its \caption in a minipage environment as shown in Fig. 2 and Table 1. Also, as in a figure environment, the caption for Table 1 is correctly typeset because the minipage for it has the \captionType{table} command to specify the type of caption. The command can of course be used with the figure argument to give a figure caption.

Figure 4 shows how to make a double column figure.

You may use any size font, as shown in Fig. 4. Also, you may include an encapsulated PostScript file (so called EPS file) as the body of a figure. To include, use

```
\usepackage{graphicx}
```

in the preamble and put the \includegraphics command where you wish to embed the EPS graphics with its file name (and options if necessary).

You might have noticed that the first reference to Fig. 1 is bold-faced, while the second and third are typed in roman fonts. This font switching is a rule of the Journal/Transactions and will be automatically performed if you use \figref{<label>} instead of Fig.~\ref{<label>}. Another rule is that “Figure” must be used instead of “Fig.” if the reference is the first word of a sentence, such as was the first reference to Fig. 4 above. Unfortunately, this switching is too hard to do automatically, so you must use \figref{<label>} in such cases.

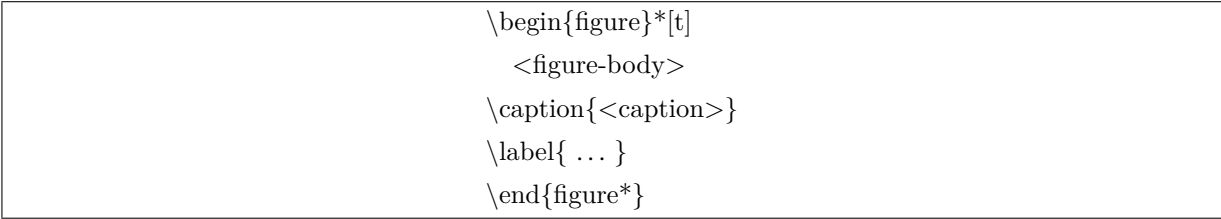


Fig. 4 Double column figure

Table 2 Sections and sub-sections in which list-like environments are used (example of table)

	enumerate	itemize	description
type-1	2	3	4.5
type-2	—	4.11	4.7
type-3	2	—	4.5
type-4	—	4.8	4.3

type-1: enumerate, etc. type-2: enumerate*, etc.
type-3: Enumerate, etc. type-4: ENUMERATE, etc.

5.8 Tables

A table with many rules is not very beautiful. Table 2 shows an example of a table with standard style rules. Note that the uppermost rule is doubled, and no rules are drawn on the left and right edges. The caption should be put above the table. The default font size for tables is `\footnotesize`. Any reference to a table should be made using `\tabref{<label>}`.

5.9 Citations, Reference, Acknowledgements

5.9.1 Citations

The command `\cite` is used to add citations in the text. Cited labels are sorted automatically and separated by using square brackets `[ ]`. Thus,

The paper `\cite{companion,latex}` is
an overview of `\LaTeX`.

will produce

The paper [1], [2] is an overview of `\LaTeX`.

5.9.2 List of References

References should be arranged in alphabetical or cited order. It is recommended to use `BiBTeX` and style files `ipsj-sort-e.bst` (alphabetical order) or `ipsj-unsort-e.bst` (cited order) to make references fit to the traditional style. Remember that you must include `.bbl` file in the file package, instead of `.bib`. If you cannot use `BiBTeX` and have to make references manually using the `bibliography` environment, observe the references of this guide carefully and follow its style.

5.9.3 Acknowledgments and Appendices

If you want to acknowledge people, put your acknowledgments just before the references and enclose them in the `acknowledgment` environment. Acknowledgments will not be printed in drafts.

Appendices, if there are any, should be put just after the references and `\appendix` command. Sectioning commands produces headings like A.1, A.2, and so on in the appendices.

5.10 Biography

Biographies of authors are positioned at the end of the document, just before `\end{document}`, as follows.

```

\begin{biography}
\profile{<1st-author's-name>}
  {<biography-of-1st-author>}
\author{<2nd-author's-name>}
  {<biography-of-2nd-author>}
.....
\end{biography}

```

6. Check List of “Dos and Don’ts”

6.1 The basics of writing

- Describe a paper so that readers understand the novelty, availability, and reliability of the research.
- Try to make a paper easy to read (discontinuity in the story and obscure backgrounds or themes are a burden to readers).
- Revisit the paper if the problem to be solved is not generalized (entirely focused on a problem at XX University, etc.) or if the paper reports deliverables only and does not describe the problem itself.
- Rethink the paper if its conclusion is not clearly described, it does not adequately point out its applicability, limits, and controversial points, or its conclusion does not follow the contents.
- Expressions that are inappropriate for scientific papers and that are hard to understand should be reconsidered.
- Second thought is necessary if sentences are in colloquial style.
- Check the structure of chapters and sections and the organization of the paper.
- Do not make the paper so that grasping the meaning is difficult without guessing from the context.
- Confirm if the explanation of the hypotheses is enough and does not contain any gaps in meaning.
- The authors should not submit a manuscript that includes redundant and/or too brief descriptions.
- The authors should eliminate undefined terminologies.

6.2 Show novelty and usefulness clearly

- The authors should not submit a manuscript that does not clarify the motivation and the goal of their study and the relationship to other existing studies.
- The authors should not submit a manuscript that does not clarify what technologies are well/publicly known

and what idea they are newly/originally proposing.

- The authors should provide sufficient references in their manuscript to back up the originality of their study.
- The authors should not submit a manuscript in which the readers cannot understand their proposal (or cannot find any originality in it) because it consists entirely of abstractive and/or conceptual descriptions.
- The authors should not submit a manuscript that lacks discussions on the effectiveness of their proposal.

6.3 Concrete attention to writing

- The authors should not submit a manuscript whose Japanese title does not match its content correctly.
- The authors should not submit a manuscript whose English title does not match its content correctly or that contains incorrect English usage.
- The paper should be revised when its abstract does not show its purpose or is written in inadequate English.
- The paper should be revised when symbols and abbreviations are not popular, wordings are not adequate, or the explanations of its pictures and tables are not adequate.
- The paper should be revised when special wordings, which are popular only in an individual or local group or a small company, are used without any explanations.
- The paper should be revised when its pictures or tables are not semantically clear or they contain mistakes.
- The paper should be revised when its pictures or tables are not visually clear.
- The paper should be revised when the size or the scale of its pictures or tables are not adequate.

6.4 Regarding references

- The number of references should be more than 10 (Some opinions say more than 20 or 30 in some research areas).
- A sufficient number of references are required to show the paper's novelty.
- The paper should be revised when it has an insufficient number of references.
- Referring to appropriate papers written by Japanese authors contributes to the further progression of the Japanese research community.
- Do not include self-citations excessively.

6.5 Double submission

- Double submission of the original paper is prohibited. However, it is permissible to submit a paper accepted at an international conference and free from copyright issues.
- Do not use the same figures or charts already included in other original papers, except those that have proper citations.
- Be careful not to have overlap between the paper and

other published articles.

6.6 Check by other researchers

- Proofreading by experienced persons with many accepted papers is strongly recommended.
- Take care to avoid leaps of logic from the viewpoint of the readers.

6.7 Miscellaneous

- After the first review round, do not modify the paper except for the stated conditions for acceptance without the reviewers' approval.
- Since the IPSJ uses a double-blind review system, in which both author(s) and reviewers remain anonymous, the authors cannot select reviewers.
- Fill the self-check sheet carefully before submitting the paper.

7. Concluding Remarks

We dare not dream that the style files are perfect but rather wish to improve them with your cooperation and hope that you will let us know of any complaints, comments, suggestions by e-mail to:

editt@ipsj.or.jp.

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Appendix

A.1. How to Write an Appendix

To add an appendix, write the command `\appendix` immediately following the reference list. Within the appendix, the `\section` command creates numbered headings such as A.1 and A.2.

A.1.1 Example of a Heading

The command `\subsection` in the appendix gives this kind of heading.

A.2. Commands for Transactions

Each transaction has its own subtitle, abbreviation code, and serial number. This information is given by using the following commands for the `\documentclass` option in the final version.

- PRO (Trans. Programming)
- TOM (Trans. Mathematical Modeling and Its Applications)
- TOD (Trans. Database)
- ACS (Trans. Advanced Computing Systems)
- CDS (Trans. Consumer Device & System)
- TBIO (Bioinformatics)
- SLDM (System LSI Design Methodology)
- CVA (Computer Vision and Applications)

Moreover, for papers in English, the command `English` can be added. For example, writing `\documentclass[PRO, english]{ipsj}` will create an English document.

Note that the research group has a “month of publication” number that does not correspond to the “issue month

number” of the transaction. You may be notified by the IPSJ or the Editorial Board of the `<month>` in order to set the month of publication counter as follows.

`\setcounter{month}{month of publication}`

In addition, commands are provided for executing unique functions for several transactions, as shown in the following sections.

A.3. Unique Commands for Each Part

Since each of the parts has its own detailed specifications, the same command may produce different results in two different parts.

In some cases, the `<Revised date>` and `<Second revised date>` are inputted. These can be added as a preamble by using

`\rereceived{<year>}{<month>}{<day>}`
`\rerereceived {<year>}{<month>}{<day>}`

A.3.1 Unique Functions for Programming (PRO)

Issues of Transactions on Programming (PRO) includes not only regular papers but also abstracts from research presentations delivered in the research groups of SIGPRO. The file for an abstract consists of material from the `\documentclass` to the of the format shown in Section 4. That is, the file does not have a main text. Note that the reception and acceptance dates are not required, but the date of presentation has to be given:

`\Presents{<year>}{<month>}{<day>}`

A.3.2 Unique Functions for Database (TOD)

The name of the editor in charge for the paper included in The Transactions on Database (TOD) is specified by `\edInCharge{<name-of-editor>}`

Also, following a change in style, the command is entered at the end of the paper, directly before `\end{document}`.

A.3.3 Unique Functions for Consumer Devices & Systems (CDS)

In the “Transactions on Consumer Devices & Systems,” since the headings differ depending on the type of document, the type of heading is to be changed with the option.

The types are:

- `systems` Paper on Consumer Systems
- `services` Paper on Consumer Services
- `devices` Paper on Consumer Devices
- `research` Research Paper

For English papers, you simply need to add English.

A.3.4 Unique Functions for Bioinformatics (TBIO)

Since papers in Transactions on Bioinformatics (TBIO) are in English, specifying the TBIO option will cause the program to assume that the English option has been specified. This effectively means that the English option can be omitted.

The following three categories define the different types of papers.

- No specification Original Paper (Default)
- Data Database/Software Paper
- Survey Survey Paper

Therefore, `\documentclass[TBIO]{ipsj}` will be an original paper, and `\documentclass[TBIO,Survey]{ipsj}` will be a survey paper.

Moreover, as with TOD, the name of the editor in charge of the paper is specified by using `\Editor`, but in this case, the text is introduced with “Communicated by.” Therefore, the name of the editor is positioned directly before `\end{document}`, as with TOD.

A.3.5 Unique Functions for Computer Vision and Applications (CVA)

The Transactions of Computer Vision and Applications is also an English language journal, allowing the English option to be omitted.

There are three classes of documents:

- No specification Regular Paper (Default)
- Research Research Paper
- system Systems Paper

As with TBIO, the name of the editor in charge is inserted, and the inserted text is introduced with “Communicated by.”

A.3.6 Unique Functions for System LSI Design Methodology (SLDM)

The Transactions of System LSI Design and Methodology (SLDM) is also an English language journal, allowing the English option to be omitted.

There are two classes of documents:

- No specification Regular Paper (Default)
- Short Short Paper

SDLM also enters the name of the editor in charge, but automatic insertion is treated differently depending on the paper.

Normally, text is inserted using “Recommended by Associate Editor;” but it is only when the “invited” option is included that the insertion text becomes “Invited by Editor-in-Chief.”